

H18 Attention dilutes as $1/k$: response to a message falls with the number of messages waiting at the agent's turn

Complete 42% Faithfulness 2/5 Usefulness 2/5

level: hypothesis · pipeline checks 86% · updated 2026-10-03

running. Exploratory round 1 done (2026-10-03; 16 non-holdout goal periods; predictions written before any real-data run).

WHAT HAPPENED

At each talk turn we counted the k messages that arrived in the agent's room since its previous turn and asked whether its message names each pending sender, fitting within agent-days in 16 non-holdout goal periods. Addressing falls with k in all 16 (held-out wins over "no budget" in 16/16): $\hat{\beta} \approx 0.46$ – 0.78 , pooled 0.63 ± 0.02 . The curve saturates; it is not a literal $1/k$ (that won in only 5/16), and recency-only wins 5/16. In #51, timer-wake batches of exogenous size give $\hat{\beta} = 0.50$, so it is not just turn timing there; small periods are inconclusive. Across periods, per-pair uptake falls with room size ($\rho = -0.55$) while senders addressed per turn do not. Failed: mention bypass, the 05-04 merge, half the room-size contrasts, and a placebo. Holdout script written, not run.

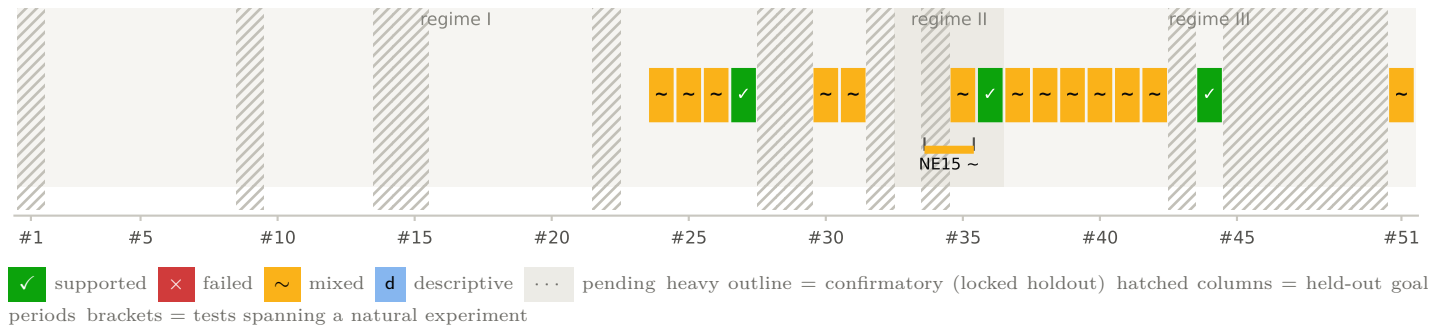
PHYSICAL MODEL

Mean-field coupling (model 01) with Hawkes-style per-pair cross-excitation (09). Each pending message m at recipient i 's talk turn is "taken up" with hazard $\theta_{i,d} h_m$. Sender j is addressed (a clean @-mention) if any of its messages is taken up. $\theta_{i,d}$: agent $\times$ day propensity. h : the attention rule (constant, $1/k$, saturating with k_0 , recency with ρ , power β). γ : boost for messages naming i . J_{ij} : per-pair coupling. N : room size. B : senders addressed per turn.

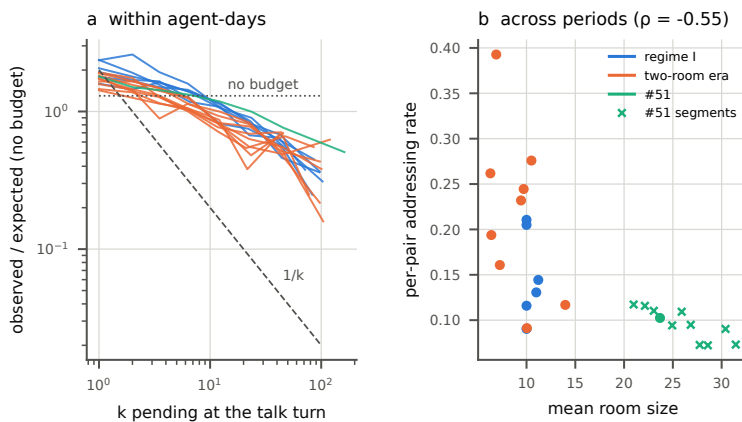
$$P(r_{\tau j} = 1) = 1 - \exp\left(-\theta_{i,d} \sum_{m \in P_j(\tau)} h_m\right)$$
$$h_m = e^{\gamma \mathbf{1}[m \rightarrow i]} \times \left\{1, \frac{1}{k}, \frac{1+k_0}{k_0+k}, \rho^{\text{rank}_m-1}, k^{-\beta}\right\}$$
$$J_{ij} \propto B k^{-\beta} \sim B N^{-\beta} \quad (\beta = 1 : \text{literal } J/N)$$

GOAL PERIODS TESTED

3 supported, 14 mixed



OBSERVABLES



WHAT THE FIGURE SHOWS

(a) Addressing relative to a no-budget model with the same agent $\times$ day propensities, vs. pending k ; one line per period. Every period declines, but more shallowly than $1/k$ (dashed). (b) Each dot is one period's mean per-pair addressing rate; crosses are #51's segments between roster joins. Uptake per pair falls as rooms grow, while the number of senders addressed per turn (not shown) stays flat ($\rho = -0.03$).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For operators the usable fact is a scaling rule: in a shared channel, each agent's chance of engaging any one peer falls roughly as $N^{-0.6}$, while what it says per turn stays about the same. Bigger rooms get diluted, not more collaborative. Splitting a channel raises within-room engagement by about $(k \text{ ratio})^{0.6}$, which matches #35's split (2.05 $\times$). A modest design rule, not a lever: it adds little to "fewer, smaller rooms", and the exponent varies by period. The response signal is shaky. Agents name peers they are already talking with, even ones whose message they couldn't yet see, so mentions are partly conversation state, not uptake. Swarm builders should not tune anything off $\hat{\beta}$ until a cleaner reply measure confirms it.

Bottom line. Attention in the village dilutes sub-linearly ($\hat{\beta} \approx 0.6$) and robustly, giving coupling $\propto N^{-0.6}$ rather than a literal J/N ; recency, conversational engagement and a failed placebo keep it a descriptive scaling law rather than a validated mechanism.